Hybrid Character Graph Extraction with Contextual Entity Filtering and Alias-Safeguarded Disambiguation

Steve B. Sanogo[1], *Malika Ghilas*[1]

[1]Avignon University, LIA, France

moussa-steve-belvin.sanogo@alumni.univ-avignon.fr, malika.ghilas@alumni.univ-avignon.fr

Abstract

This paper presents a hybrid pipeline for extracting polarity-aware character networks from French literary narratives. Character graph extraction from fiction remains challenging because named entity recognition is noisy on literary text, character mentions are fragmented across aliases and titles, co-occurrence is only an approximate proxy for interaction, and relation polarity is rarely modeled explicitly. We address these issues with a modular pipeline combining Flair and SpaCy named entity recognition, PER-priority label aggregation, contextual entity filtering, alias resolution with merge safeguards, weighted co-occurrence graph construction, and chapter-level ternary polarity assignment. The method is evaluated on 37 chapters from two Asimov novels, *The Caves of Steel* and *Prelude to Foundation*. The best configuration reaches $F_1 = 0.664$ with a 20-token co-occurrence window. Experiments show that window calibration is critical, while global edge-weight filtering is overly destructive. The extracted centralities remain consistent with the main narrative protagonists.

Index Terms: character networks, literary NLP, named entity recognition, alias resolution, polarity analysis

1. Introduction

The automatic extraction of character networks from narrative texts is a core problem in literary NLP and digital humanities. Once extracted, a character network exposes the interaction structure of a text and supports downstream analyses of role, centrality and narrative dynamics [1, 2]. Building such networks at scale requires reliable named-entity recognition, fine-grained entity resolution, and an interaction model that goes beyond raw co-occurrence.

These three requirements are difficult to satisfy on literary prose. Generic named-entity recognisers degrade on long, stylistically rich documents whose vocabulary departs from news-domain training data; characters surface under multiple aliases, titles and short or long forms that fragment the entity set; and the standard window-based co-occurrence signal is at best an imperfect proxy of relation, since many narratively important interactions are implicit, indirect, or realised through dialogue and free indirect speech rather than through direct mention. Polarised relations (alliance, antagonism) are especially elusive on this kind of text and have only been partially addressed by previous work.

We address these issues with a hybrid pipeline applied chapter by chapter to a 37-chapter subset of French translations from Asimov's Robots/Foundation universe (18 chapters of *The Caves of Steel*, 19 of *Prelude to Foundation*). The main contributions of this work are: (i) a hybrid NER pipeline combining Flair and SpaCy with a PER-priority voting scheme; (ii) a contextual entity filtering strategy based on lexical human-likeness scoring; (iii) an alias resolution and structural graph disambiguation module with explicit safeguards against unsafe merges; (iv) a chapter-level ternary polarity model labelling pairs as *ami*, *neutre* or *ennemi*; and (v) an experimental analysis of the co-occurrence window and edge-filtering hyperparameters that drive the pipeline. The final configuration reaches $F_1 = 0.664$ at co-occurrence window $W = 20$ without any global edge-weight filter.

2. Related Work

Character networks from fiction. The extraction and analysis of character networks from narrative texts is the subject of a sizeable body of work, surveyed by Labatut and Bost [1]. Most pipelines decompose the task into three stages: identifying characters, detecting their interactions and assembling a graph. Earliest methods built undirected co-occurrence networks at the level of paragraphs, scenes or fixed-token windows [2, 3], an approach that captures macro-structure (protagonists, sub-arcs, communities) but provides no information about the type of relation. Subsequent modular pipelines have refined the individual stages while keeping the underlying interaction signal essentially co-occurrence-based [1].

NER and entity resolution in literary text. Off-the-shelf NER models trained on news data degrade markedly on literary prose, and the long, closed-world nature of novels amplifies this gap by surfacing the same characters under many lexical variants (first name, family name, title, nickname). Recent work argues that contextual cues, both local and document-level, are critical to recover these mentions: Amalvy et al. show that augmenting a tagger with retrieved global context improves NER on book-length documents [4], and propose a learning-to-rank approach that selects relevant context windows from a synthetically generated dataset [5]. Our hybrid Flair/SpaCy combination follows the same intuition by trading single-model precision for cross-model recall, then restoring precision through a contextual filter and a conservative alias map.

Polarity-aware character graphs. Co-occurrence by itself encodes only that two characters are textually close, not the nature of their interaction. Extending character networks with a relational sign or polarity has been comparatively under-studied [1]: a few works enrich edges with affective scores, and signed-network analyses have been applied post-hoc, but few systems jointly extract characters *and* a typed, polarised interaction at the chapter level. We contribute a lightweight ternary polarity classifier (*ami* / *neutre* / *ennemi*) operating per chapter and per character pair on top of the extracted graph, and analyse its behaviour against the topological structure in Section 4.

3. Methodology

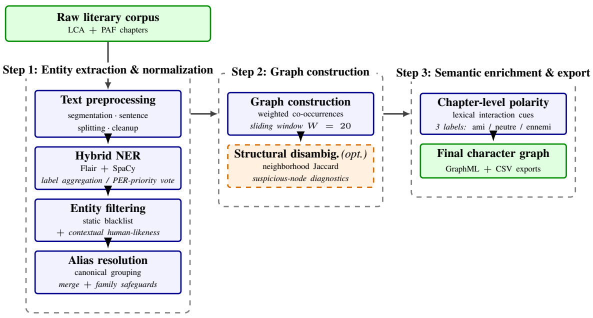

Figure 1: *Overview of the proposed pipeline for polarity-aware character graph extraction.*

3.1. Pipeline Overview

We process each book of the corpus chapter by chapter. A chapter is taken through seven stages: (i) text preprocessing (cleaning, sentence and token segmentation); (ii) hybrid named-entity recognition combining Flair and SpaCy; (iii) entity filtering, made of a static blacklist and a contextual scorer; (iv) alias resolution, which groups surface variants under a canonical form; (v) weighted co-occurrence graph construction with a sliding window; (vi) optional structural disambiguation based on neighbourhood similarity, used as a diagnostic safeguard rather than as an aggressive merger; and (vii) chapter-level relational polarity, which labels each pair as *ami*, *neutre* or *ennemi*. The chapter-level graphs and per-pair polarity decisions are exported as GraphML and CSV, and aggregated into per-book networks for downstream analysis. The same pipeline is applied without modification to the two sub-corpora considered in this work (*The Caves of Steel*, LCA, and *Prelude to Foundation*, PAF), so that any difference observed in Section 4 can be attributed to the texts themselves and not to corpus-specific tuning.

3.2. Hybrid Named Entity Recognition

The input chapter is processed in parallel by two NER engines: a Transformer-based Flair tagger and a SpaCy statistical model. Each engine yields a set of typed mentions (surface, type, span). We aggregate the two outputs at the mention level using a span-overlap voting scheme. When two predictions cover the same surface form but receive different labels, the PER label is given priority over PER-incompatible labels (LOC, ORG, MISC), since the goal of the pipeline is to maximize the recall of character mentions before downstream filtering. Mentions predicted by a single engine are retained for character graph construction only when their type is PER. This hybrid strategy deliberately favours recall at the NER stage; precision is then restored downstream by the contextual filter (Section 3.3).

3.3. Contextual Entity Filtering

NER false positives are removed in two steps. A static blacklist first discards lexically obvious non-character mentions (toponyms, generic common nouns mistakenly capitalised, narrator artefacts, recurring formulaic phrases). The remaining candidates are then scored by a contextual filter whose role is to verify that an entity actually *behaves* as a person in its local textual context.

For an entity e with occurrence set $\mathcal{O}(e)$, we look at the local token context C_o of each occurrence o, using a symmetric window around the mention. Each verb v and each adjective a in C_o is scored against two manually curated lexicons: a *human* lexicon (verbs of speech, motion, perception, emotion; adjectives describing personhood) and a *non-human* lexicon (verbs of geographical, mechanical or institutional action; adjectives describing places or objects). Scoring functions $\phi_v(\cdot)$ and $\phi_a(\cdot)$ return positive values for human markers and negative values for non-human ones. The contextual score of the entity is the average over its occurrences:

$$s(e) = \frac{1}{|\mathcal{O}(e)|} \sum_{o \in \mathcal{O}(e)} \left(\sum_{v \in C_o} \phi_v(v) + \alpha \sum_{a \in C_o} \phi_a(a) \right), \quad (1)$$

where α down-weights the adjective channel relative to the verb channel. In our experiments, the symmetric context window spans 10 tokens on each side of the entity, the adjective weight is $\alpha = 0.7$, and the acceptance threshold is $\tau = -1.5$. Entities with $s(e) < \tau$ are discarded; zero-score entities are kept by default to avoid over-filtering sparse mentions. All lexical resources (blacklist, human/non-human contextual markers, positive and negative relational verb lexicons) were manually curated from corpus error analysis and kept fixed across both books; they are made available with the experimental scripts.

3.4. Alias Resolution and Structural Disambiguation

Literary characters surface under several textual forms (first name, family name, title, nickname, long versus short form). We build, for each chapter, an alias map in which textual variants are gathered into groups, the first element of each group being the canonical form selected for the graph. Co-occurrence counting is performed on the original surface forms and then attributed to the canonical node, so that no occurrence is lost and the alias map only relabels nodes. A small set of safeguards prevents unsafe merges, in particular between distinct characters that share a family name or a generic title (*Dr*, *Lieutenant*); such candidates are kept apart unless additional evidence supports the merge.

A second, complementary mechanism operates after the graph is built. For each pair of nodes (u, v) we compute the Jaccard similarity of their neighbourhoods,

$$J(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}, \quad (2)$$

where $N(\cdot)$ denotes the set of co-occurring nodes. Pairs with high $J(u, v)$ are flagged as suspect and surfaced for inspection. We use this *structural disambiguation* module conservatively, primarily as a diagnostic and validation tool: it highlights residual alias issues that the lexical map missed, but in the final pipeline its automatic merges are kept off so as not to risk merging genuine narrative twins (for instance two recurring antagonists who share most of their interlocutors).

3.5. Graph Construction

Per chapter c, the canonical entities form the node set, and a weighted edge is created between two characters whenever a mention of one falls within W tokens of a mention of the other. The chapter-level edge weight is therefore

$$w_{ij}^{(c)} = \sum_{m_i \in M_i^{(c)}} \sum_{m_j \in M_j^{(c)}} \mathbf{1}\left[|\mathrm{pos}(m_i) - \mathrm{pos}(m_j)| \le W\right], \quad (3)$$

where $M_i^{(c)}$ and $M_j^{(c)}$ denote the mention sets of characters i and j in chapter c, and $\mathrm{pos}(\cdot)$ is the token position of a mention. The final window is $W = 20$ tokens, a value selected on the development set (Section 4.3). Isolated nodes are pruned before export. The resulting per-chapter graphs are written as GraphML, together with a CSV of edges and per-pair polarity. Aggregated per-book networks are obtained by summing edge weights over the chapters of the book and re-pruning isolated nodes.

3.6. Chapter-Level Polarity Analysis

Relational polarity is estimated per chapter and per character pair. We score each sentence with two manually curated lexicons of relational verbs: a positive lexicon (verbs of friendship, support, protection, trust) and a negative lexicon (verbs of conflict, attack, betrayal, threat). Negation in the immediate neighbourhood of a marker inverts its sign. Whenever two characters co-occur in a sentence containing a signed marker, the corresponding signal is attributed to their pair. The chapter-level polarity of a pair is then the majority vote over its collected signals: a positive majority yields the *ami* label, a negative majority the *ennemi* label, and the absence of signal or a tie defaults to *neutre*. The output label set is therefore ternary: $\{\mathrm{ami}, \mathrm{neutre}, \mathrm{ennemi}\}$.

This lexicon-and-vote scheme is, in our setting, more robust than a strict subject–verb–object parser: in literary French, relations between characters are frequently expressed indirectly (free indirect speech, pronominal subjects, ellipsis, focalised narration), which makes SVO recovery brittle and propagates parser errors into the polarity labels. By relying on co-presence in the same sentence and on signed lexical markers, we trade fine-grained semantic parsing for a more stable signal that degrades gracefully on the narrative passages typical of Asimov's prose.

4. Experiments and Results

4.1. Dataset and Evaluation Protocol

We evaluate the pipeline on French translations of a 37-chapter subset of two Asimov novels, made of *The Caves of Steel* (LCA, 18 chapters) and *Prelude to Foundation* (PAF, 19 chapters). Due to source-availability constraints, only roughly half of the original pages were available for processing. We treat this as a property of our experimental setup, since it introduces discontinuities in entity mentions, alias evidence and local interaction contexts. The pipeline is run chapter by chapter and produces, for each chapter, a weighted signed graph whose nodes are canonical character names and whose edges are typed by the ternary polarity scheme of Section 3.6 (*ami*, *neutre*, *ennemi*).

The benchmark scores each predicted $(chapter, character\ \ pair, polarity)$ triple against the manually annotated reference. Character pairs are treated as undirected after canonicalisation. A prediction is counted as correct only when both the canonical pair and the ternary polarity label match the reference. We report micro-F_1 over all evaluated triples. The best configuration reaches $F_1 = 0.664$ at co-occurrence window $W = 20$ tokens, with no global edge-weight filter. For reproducibility, the source code, experimental scripts, regression tests and a live demo are available from the project repository[1].

[1] `https://github.com/steve-sanogo/RxPersonna`

4.2. End-to-End Pipeline Improvement

As a coarse reference point, the initial hybrid pipeline already combined Flair and SpaCy NER, a static blacklist, text-based alias resolution and a legacy polarity module based on SVO patterns, dialogue cues and affiliation signals. This initial configuration reached approximately micro-$F_1 = 0.57$ on the same benchmark. The final pipeline reaches 0.664 after adding contextual entity filtering, graph-based diagnostic disambiguation, enriched blacklist corrections, and targeted alias safeguards. We do not report a full component-level ablation, since intermediate variants were not all collected under a unified evaluation protocol. Instead, we analyse the two hyperparameters that most directly affect the final graph: the co-occurrence window and the global edge-weight threshold.

4.3. Co-occurrence Window Analysis

The co-occurrence window W is the most sensitive hyperparameter of the pipeline. A sweep on the development set (Table 1) describes a clear inverted-U: small windows ($W \leq 15$) starve the graph of relational evidence by missing interactions spread over a wider local context, while large windows ($W \geq 25$) inflate it with spurious co-mentions between characters that are merely listed together in expository paragraphs. The optimum is reached at $W = 20$ ($F_1 = 0.664$), and the score drifts down by 1–2 points over $W \in [25, 40]$. The shape of the curve confirms that co-occurrence has to be calibrated finely: it is neither a parameter that can be set blindly to a sentence boundary nor one that benefits from large-context heuristics.

Table 1: *Effect of the co-occurrence window size on extraction performance (Experiment A; final W in bold).*

W (tokens)	F_1
10	0.599
15	0.633
20	**0.664**
25	0.658
30	0.650
35	0.651
40	0.647

4.4. Edge Filtering Analysis

We also evaluated a global edge-weight filter that drops every edge with weight below a threshold $w_{\min}$ (Experiment B, Table 2). The resulting curve is monotonically decreasing and steep: dropping edges of weight 1, which would intuitively prune co-occurrence noise, in fact destroys 12 F1 points, and any further threshold collapses the score below 0.40. The reason is that the long tail of low-weight edges is not pure noise: many narratively pivotal pairs (single but decisive scenes, brief antagonistic encounters) share only a handful of tokens of textual proximity. Co-occurrence weight is therefore a poor proxy of relational salience, and the final pipeline keeps $w_{\min} = 1$. This is a deliberately negative result: it shows that pruning has to be informed by structural or semantic criteria rather than by raw weight.

Table 2: *Effect of a global minimum-edge-weight filter at $W = 20$ (Experiment B; final $w_{\min}$ in bold).*

$w_{\min}$	F_1
1	**0.664**
2	0.544
3	0.360
4	0.345
5	0.281

4.5. Graph-Level Analysis and Polarity

We finally summarise the structure of the networks produced by the best configuration (Table 3). Aggregating the chapter-level graphs into a single weighted graph per book yields 28 nodes / 57 edges for LCA and 51 nodes / 123 edges for PAF. Both networks display the expected signatures of social/narrative graphs: a short diameter (4), a near-complete giant component, and a heavy-tailed weighted-degree distribution dominated by one or two protagonists. The top-ranked characters by weighted degree match the canonical foreground of each novel: Baley ($s = 265$) and Daneel ($s = 196$) dominate LCA, while PAF is centred on Hari Seldon ($s = 424$), with a 4× gap to the next tier (Dors Venabili, Raych, Hummin, Cleon).

The lower part of Table 3 reports the distribution of the ternary polarity labels over the 332 chapter-level edges. The neutral class dominates strongly in both books (87.8% in LCA, 93.8% in PAF). Figure 2 makes this imbalance explicit: most extracted chapter-level relations are neutral, whereas positive and negative relations form a small but informative minority.

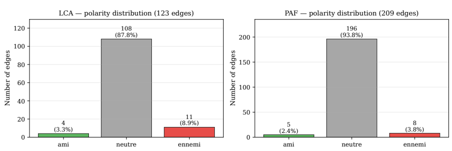

Figure 2: *Distribution of chapter-level relation polarities in the final extracted graphs. Neutral relations largely dominate in both books, while positive and negative links remain comparatively sparse.*

The system does, however, recover a small but non-empty set of antagonistic edges (11 in LCA, 8 in PAF), confirming that ternary lexical scoring is sufficient to surface the confrontational episodes (Baley – Clousarr in LCA, conflictual interactions around Seldon in PAF) that a purely co-occurrence-based extractor would otherwise flatten to neutral. The asymmetry $|\text{ennemi}| > |\text{ami}|$, the dominance of the neutral class, and the implications of these phenomena for relational interpretation are discussed in Section 5.

Table 3: *Structure of the aggregated networks and chapter-level polarity distribution at the final operating point ($W = 20$, $F_1 = 0.664$).*

	LCA	**PAF**
Chapters	18	19
Aggregated nodes	28	51
Aggregated edges	57	123
Avg. weighted degree	29.1	26.5
Diameter (LCC)	4	4
Avg. clustering	0.57	0.64
Chapter-level edges by polarity		
ami	4 (3.3%)	5 (2.4%)
neutre	108 (87.8%)	196 (93.8%)
ennemi	11 (8.9%)	8 (3.8%)
Total	123	209

5. Conclusion and Future Work

We have presented a hybrid pipeline for character-graph extraction from literary narratives, combining a Flair+SpaCy named-entity recogniser with a PER-priority vote, a contextual filter based on lexical human-likeness, an alias map with safeguards against unsafe merges, a windowed co-occurrence graph builder, and a chapter-level ternary polarity classifier (*ami*/*neutre*/*ennemi*). On the 37-chapter Asimov subset, the final configuration reaches $F_1 = 0.664$ at co-occurrence window $W = 20$ tokens. The hyperparameter sweeps further show that this window must be calibrated finely and that a global minimum-edge-weight filter, although appealing, is too destructive: it collapses recall and is therefore not used in the final pipeline.

The main residual limitations are the incomplete source material, the strong dominance of the neutral class in the polarity output, the inherent weakness of co-occurrence as a relational proxy, the dependence on manually curated lexicons, and the absence of full neural coreference. Promising directions include adaptive co-occurrence windows, semantic embeddings for entity disambiguation, neural coreference, and LLM-assisted relation extraction. Finally, future work could extend the polarity module by combining SVO patterns, dialogue cues and affiliation signals into a continuous score, then applying chapter-level Softmax normalisation instead of fixed global thresholds.

6. References

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